

Recap

Caching is an important mechanism implemented within the DNS to speedup lookup phases

Effectiveness is measured in terms of hit/miss rates

Caches store all resource records included in the response messages

Fewer RTTs to obtain the responses

Reverse mapping is used to obtain names from IP addresses

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DNS security

DNS is an open distributed service often seen as a utility whose main design principles were availability, reliability and efficiency of the service rather than security

DNS is a critical infrastructure and outages due to security incidents might lead to devastating effects

DNS is robust: during its long lifetime it survived many massive security attacks (with some disruptions of the service), e.g., “large” DDoS attack against Root Name Servers in 2002, huge DDoS attack of Mirai botnet against Dyn DNS provider in 2016

DDoS attack that disrupted internet was largest of its kind in history, experts say

Dyn, the victim of last week's denial of service attack, said it was orchestrated using a weapon called the Mirai botnet as the 'primary source of malicious attack'

Major cyber attack disrupts internet service across Europe and US

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Major DDoS attack on Dyn disrupts AWS, Twitter, Spotify and more

Well this is not good

October 21, 2016 By: Sebastian Moss Comment

Dyn DDoS Attack Highlights Vulnerability of IoT Devices

What can businesses do to guard against vulnerabilities as they bring more connected devices into their IT environments?

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Types of security risks

The security risks affecting DNS services can be classified in two categories:

- Risks affecting *infrastructure* (i.e., Name Servers) whose main goal is to make the services unavailable
- Risks affecting *data* (i.e., DNS database) whose main goal is to divert traffic towards malicious servers by providing fake Resource Records in the response messages

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DDoS attack vectors

DDoS attacks aim at exhausting Name Server resources and/or saturating the network

DDoS attacks exploit different techniques:

- Flooding the target Name Server (e.g., TCP SYN flood, UDP flood, NXDOMAIN attacks)
- Resolver and zone file vulnerabilities, such as unpatched software, Name Server misconfigurations (e.g., TsuNAME vulnerability with cyclic dependencies)
- Amplification techniques (i.e., small queries with “big” response messages) to make DDoS attacks more aggressive and effective

Reflection is typically used in the attacks, i.e., spoofed IP address to hide the actual source of the queries

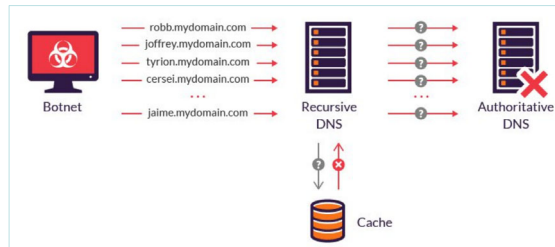
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NXDOMAIN attacks

Objective is to disrupt an Authoritative Name Server by exploiting a recursive resolver and a botnet



Laundering attacks are based on a slow and methodical process very damaging in the long run (DNS water torture)

Subdomain prefixes are continuously randomized and used only once

CPU overload and cache pollution of DNS resolver & overload of the Authoritative Name Server

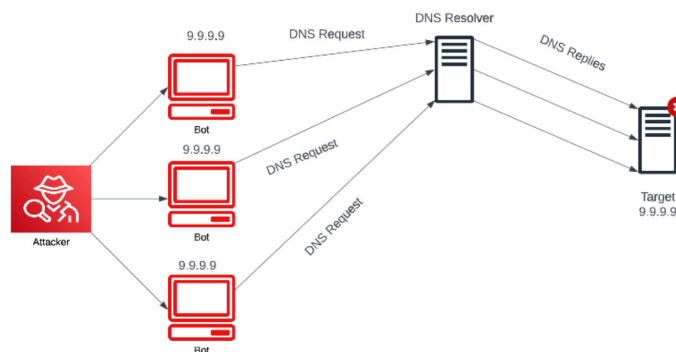
Difficult to detect and mitigate because queries originate from a legitimate Name Server and the targeted domain is genuine

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Reflection/amplification techniques



Bandwidth is saturated by big response messages (e.g., query of type **ANY**)

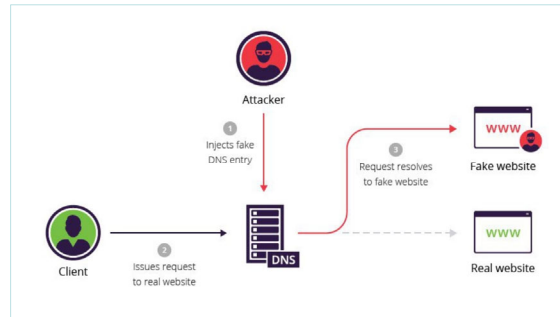
Name Servers appear to be under mutual attack

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Security risks affecting DNS database



There are multiple ways for intercepting and rerouting, or impersonating traffic on the Internet, so that the answer to a DNS query is fake

Malicious Name Servers deliver fake answers to DNS queries

Major impact on DNS caches (cache poisoning)

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Security risks affecting RRs

Two main types of attacks against DNS database content

- *On path attacks*: the idea is to intercept the 16 bits query ID & source port number by eavesdropping the traffic and then race to generate a fake response message with the objective of diverting traffic towards malicious websites/services

```
> User Datagram Protocol, Src Port: 54347, Dst Port: 53
  > Domain Name System (query)
    > Transaction ID: 0xa3ae
    > Flags: 0x0100 Standard query
      Questions: 1
      Answer RRs: 0
      Authority RRs: 0
      Additional RRs: 0
    > Queries
      > p-ring.msedge.net: type A, class IN
```

- *Hijacking attacks*: the idea is to infect devices with malware with the aim of modifying their DNS settings and have them pointing to rogue Name Servers whose answers redirect users towards malicious websites/services

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Security requirements

DNS is a critical infrastructure not designed to be secure, so it is the ideal target for security attacks

Attackers can modify results of DNS queries with fake IP addresses

It is important to ensure the integrity of the DNS lookup results (i.e., correctness and authenticity of the IP addresses returned in the response messages)

Some security problems have been fixed

The Berkeley Internet Name Domain (BIND) forbids anything not related to the original request to be cached and rejects all the “out of zone” data

It is necessary to use mechanisms able to ensure that client resolvers can trust the answers to DNS queries

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DNSSEC

One solution to fight counterfeit DNS answers is to add data origin authentication and data integrity assurance to the Domain Name System using digital signatures based on public key cryptography

DNSSEC refers to standardized DNS security extensions, that is, a hierarchical distributed trust system to validate the mappings of names to values [RFC 4033 “DNS Security Introduction and Requirements” published in 2005]

DNS Resource Records of a given domain are signed by the domain owner to allow the verification (i.e., proof of the identity of the Name Server that generates the answer) without having to trust the network or resolvers involved

DNS queries and responses are not cryptographically signed

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Security extensions

DNSSEC extensions provide:

- a) origin authentication of DNS data allows a resolver to cryptographically verify that the received data comes from the zone where it believes the data originated
- b) data integrity protection allows the resolver to know that the data has not been modified in transit since it was originally signed by the zone owner with the zone private key
- c) authenticated denial of existence allows the resolver to validate that a certain domain name does not exist or to signal that a domain name exists, but does not have the specific RR type [RFC 7129 “Authenticated Denial of Existence in the DNS” published in 2014]

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DNSSEC implementation

DNSSEC implementation leverages some specific Resource Record types (e.g., **DNSKEY**, **RRSIG**, **NSEC**)

A public/private key pair is associated with each zone

The zone public key (signed using the parent zone private key) is published in the DNS database in the **DNSKEY** record [offline]

The zone owner uses its private (secret) key to sign DNS Resource Record sets and publishes the digitally signed **RRSIG** records in the DNS database [offline]

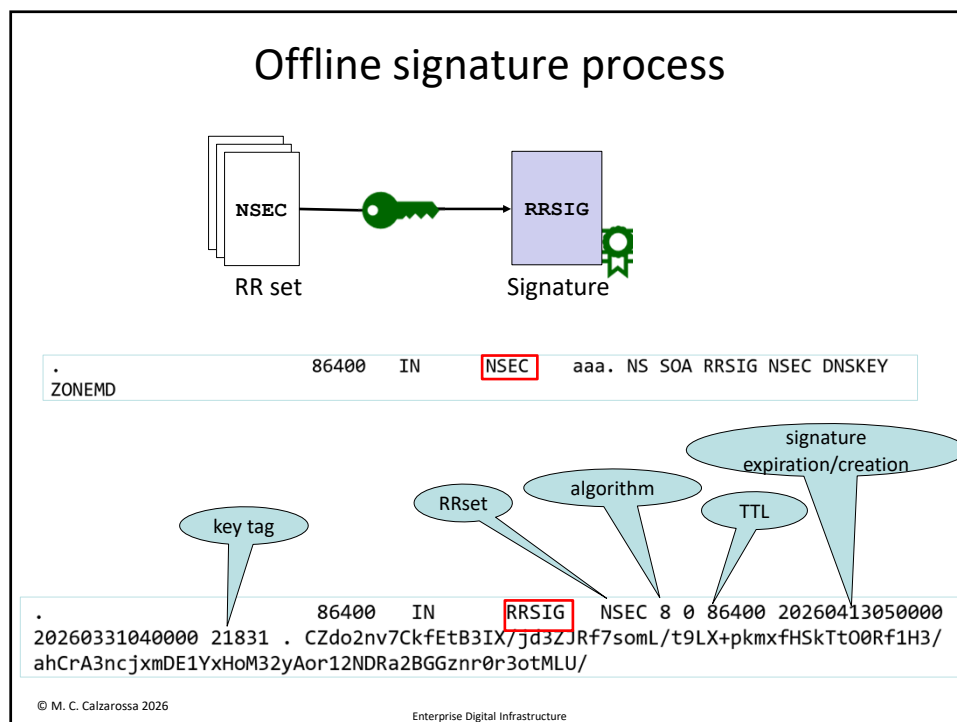
Resolvers assess the authenticity of the DNS RRs using the zone public key

- Digital signature of RR set valid: answer is legitimate and the resolver returns it to the client
- Digital signature of RR set not valid: the resolver discards the data and returns an error to the client

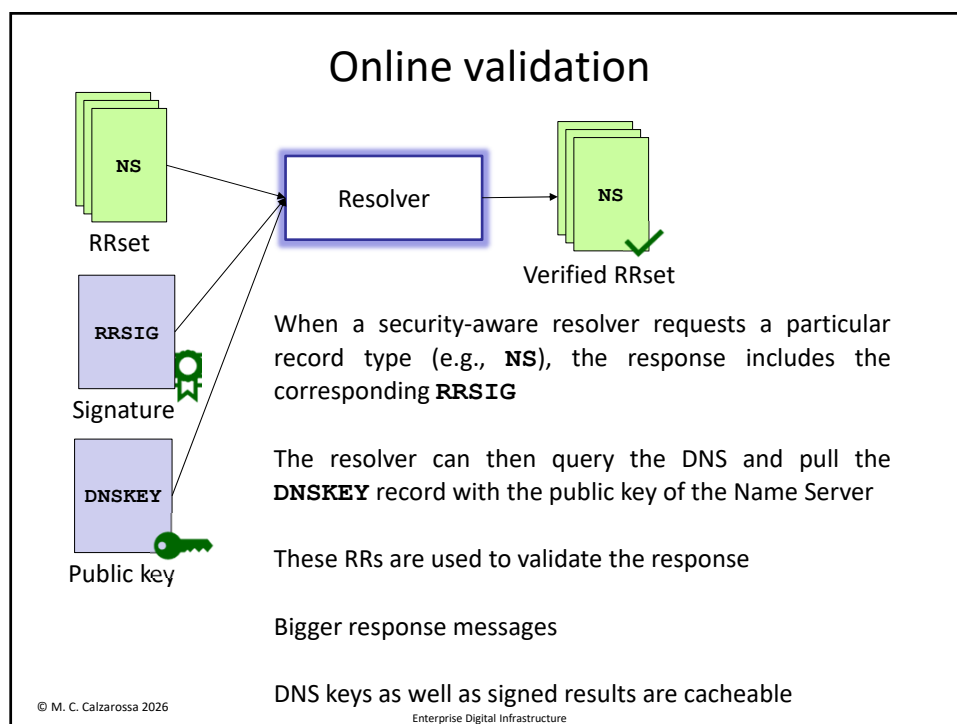
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